

Sub: _____

Date: ____/____/____

Solution: 1

(i) Host bits = $32 - 17 = 15$

$\therefore$ total number of hosts = $2^{15} - 2 = \boxed{32766}$

(ii) $\boxed{32.99.127.255}$

(iii) Network address = $32.99.0.0/17$

Sub-networks	IP needed	$2^n \geq \text{IP needed}$ n	num. of host bits	prefix	Network address
LAN-A	2002	2^{11}	11	$32-11 = 21$	$32.99.0.0/21$
LAN-B	1026	2^{11}	11	$32-11 = 21$	$32.99.8.0/21$
LAN-C	5	2^3	3	$32-3 = 29$	$32.99.16.0/29$
WAN 1	4	2^2	2	$32-2 = 30$	$32.99.16.8/30$
WAN 2	4	2^2	2	$32-2 = 30$	$32.99.16.12/30$

$(\text{ex p2d1}) - (\text{ps-isp2}) \text{ ex p2d1} =$

Solution : 2

- (i) The 'Port Forwarding' feature must be configured.
- (ii) The network is using PAT because PAT allows users to use the internet simultaneously under a single public IP while differentiating them by their port addresses.

Solution : 3

- (i) Number of fragments = $\frac{6421 - 24}{1624}$
- $= 3.94$
- $\approx \boxed{4}$
- (ii) Data size of the last packet

$$= 1624 \times 3 (6421 - 24) - (1624 \times 3)$$

$$= 12 \cdot \boxed{1525 \text{ bytes}}$$

Sub: _____

Date: ____ / ____ / ____

(iii) Bytes before last fragment = 1624×3
 $= 4872$

$\therefore$ Offset of the last fragment = $\frac{4872}{8} = \boxed{609}$

(iv) The destination uses Identification number (5656) of and fragment offset of each fragment to reassemble the fragments serially.

(v) When $MF = 1$; it indicates that there are more fragments yet to come $\frac{y}{y}$

and when $MF = 0$; ~~generally~~ mainly the MF of the last fragment is 0 which means there is no fragment after this one.

Sub: _____

Date: ____/____/____

Solution : 4

(i) ip route 0.0.0.0 0.0.0.0 ~~10~~¹⁵ 10

(ii) ip route 0.0.0.0 0.0.0.0 192.168.12.193 15

Solution : 5

R1 and R4 will converge faster because they are running LSR.

→ DVR keeps track of ^{directly} ~~directly~~ connected neighbours.

→ LSR keeps full neighbour adjacency database.

Sub: _____

Date: / /

Solution: 6

- (i) 2001 : db8 : 85a3 :: 8a2e : 370 : 7334
- (ii) 2607 : 0 : 0 : 805 : 200e
- (iii) Same as the original.

Solution: 7

- (i) Source MAC : PC A's MAC
Destination MAC : Broadcast (FF : FF : FF : FF : FF : FF)
- (ii) R1 will get the ARP request and forward to PC D.
- (iii) PC A will update its ARP cache with PC D's MAC-IP mapping and then send data packets.

Sub: _____

Date: ____/____/____

Solution: 9

(i)

- When current lease reaches 50%, client sends DHCP request.
- The DHCP server replies.
- Lease gets renewed.

(ii) If the renewal fails ^{when} at lease is at 50%, the client sends another ^{DHCP} request at 87.5% of lease. And if the client doesn't get its IP ~~at~~ before the lease expires, it drops the IP and keeps broadcasting DHCP requests.

Sub: _____

Date: ____/____/____

Sat	Sun	Mon	Tue	Wed	Thu	Fri
☐	☐	☐	☐	☐	☐	☐

Solution: 10

Loopback address is used for testing TCP/IP stack, internal communication and diagnostics.

IPv4 loopback : 127.0.0.1

IPv6 loopback : & ::1

Solution: 11

(i)

- R2 receives the ~~frame~~ packet
- checks destination MAC
- ~~finds dest~~ forwards the packet to F2 port of S3.

(ii) We call switches to be self-learning because

- Switch learns source MAC and incoming port automatically.
- Builds MAC table dynamically.
- No manual configuration needed.